EVALUATING THE SIGNIFICANCE OF OUTDOOR ADVERTISING UTILIZING COMPUTER VISION

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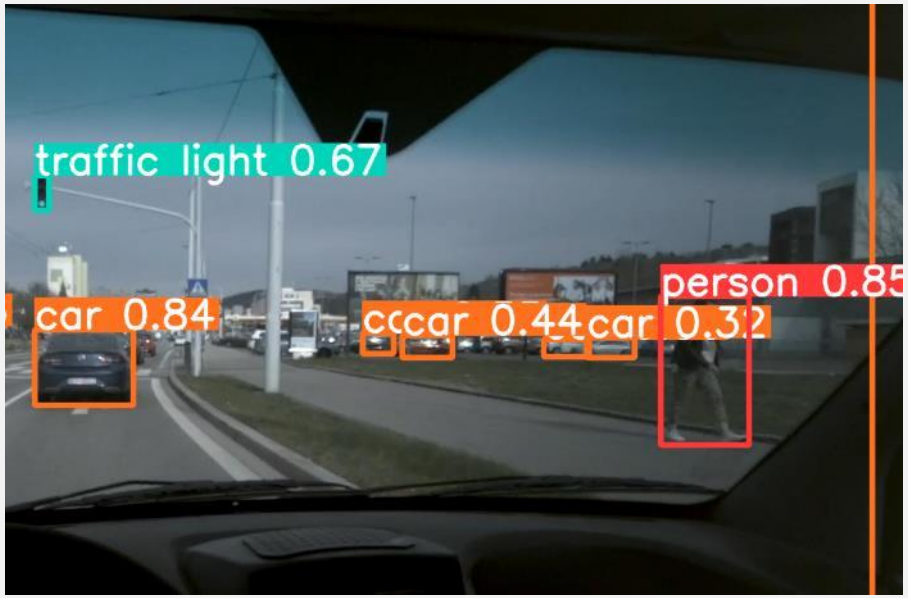
MOTIVATION

- How long have you stared at the advertisement?
- Impact of advertisement on drivers.
- Contribution to road safety.
- An application of Deep Learning in Computer Vision.

GOALS



- Long-term goals:
  - Develop an automated system which evaluates road advertisement significance.
  - Object detection, tracking, deep learning.
- Short-term goals:
  - Acquire fundamental knowledge on object detection and deep learning.
  - Train successfully an object detector that will be part of the final pipeline.

DATASET

- BillboardLamac Dataset
  - Existing dataset of 8 videos captured by eye tracker worn by 8 different drivers.
  - 145 unique billboards with 155912 bounding boxes.
  - Annotated billboards with unique ID consistent in all videos.
- Object detector
  - Train/validation/testing: 794/199/219 images.

SUPPLEMENTARY DATASET

- Mapillary Vistas Dataset
  - Large scale street level image dataset.
  - 25000 high-resolution images.
  - 66 classes (billboards included).
- Object detector
  - Images without billboards filtered.
  - Train/validation/testing: 2000/500/500 images.

OBJECT DETECTION: YOLO

- You Only Look Once (YOLO):
  - Series of Convolutional Neural Network models designed for real-time object detection.
  - Developed on PyTorch framework.
  - Detection models pre-trained on COCO dataset.
  - Classification models pre-trained on ImageNet dataset.

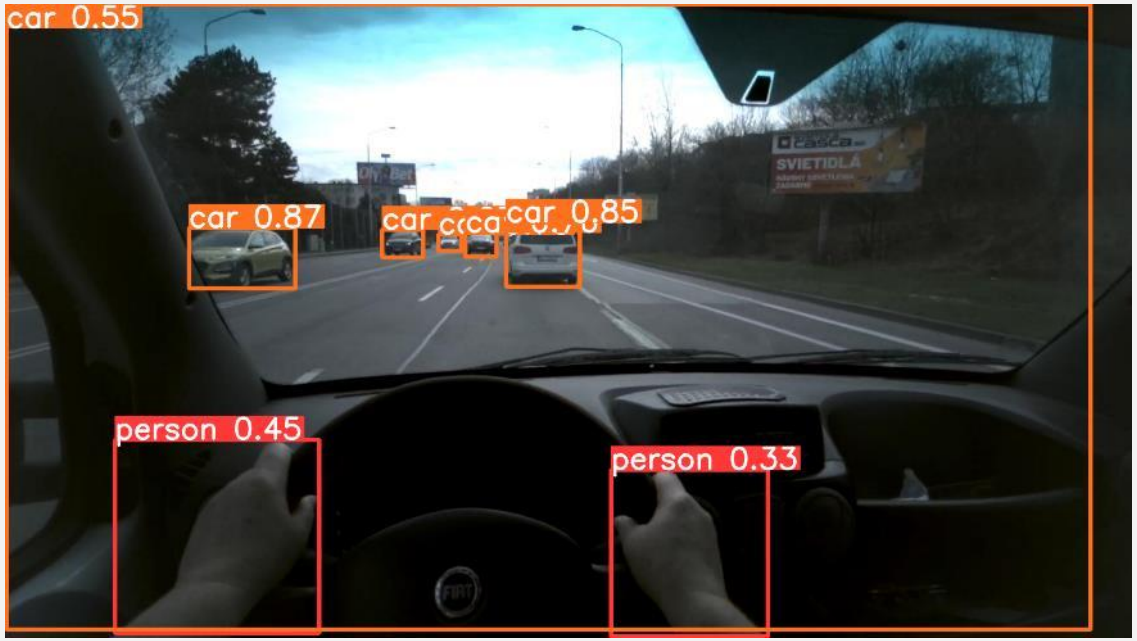
ultralytics YOLOv8

Model	size (pixels)	mAP^{val} 50-95	params (M)
YOLOv8n	640	37.3	3.2
YOLOv8s	640	44.9	11.2
YOLOv8m	640	50.2	25.9
YOLOv8l	640	52.9	43.7
YOLOv8x	640	53.9	68.2

OBJECT DETECTION RESULTS: 0

MAPILLARY VISTAS TEST SET

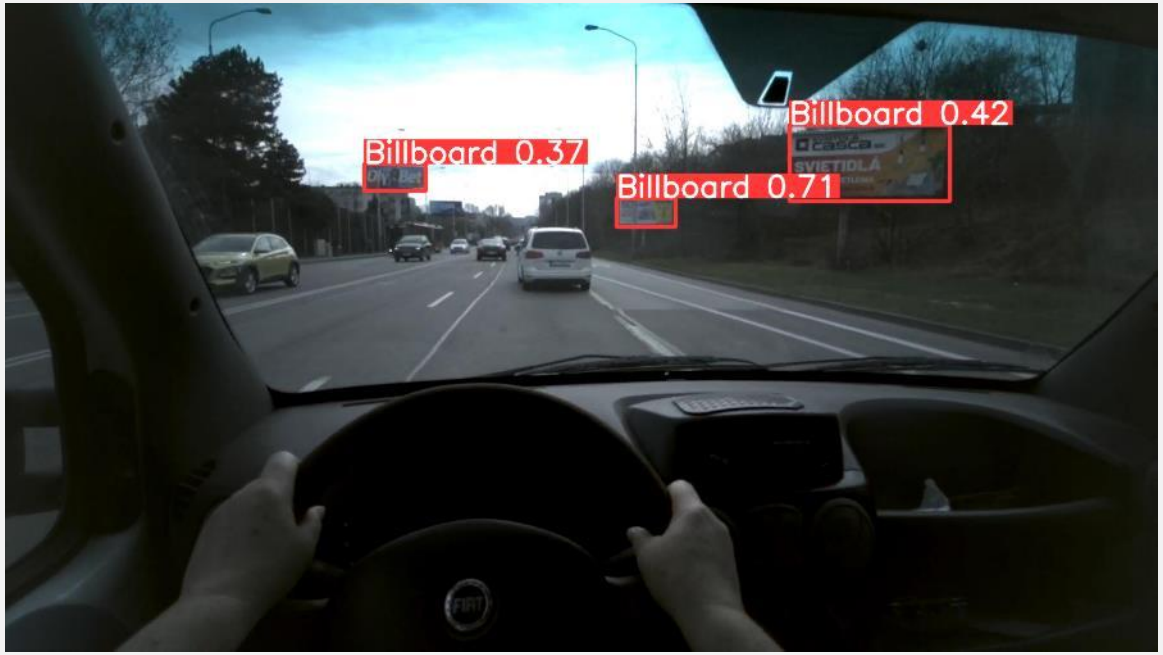
BILLBOARDLAMAC TEST SET



OBJECT DETECTION RESULTS: I

MAPILLARY VISTAS TEST SET

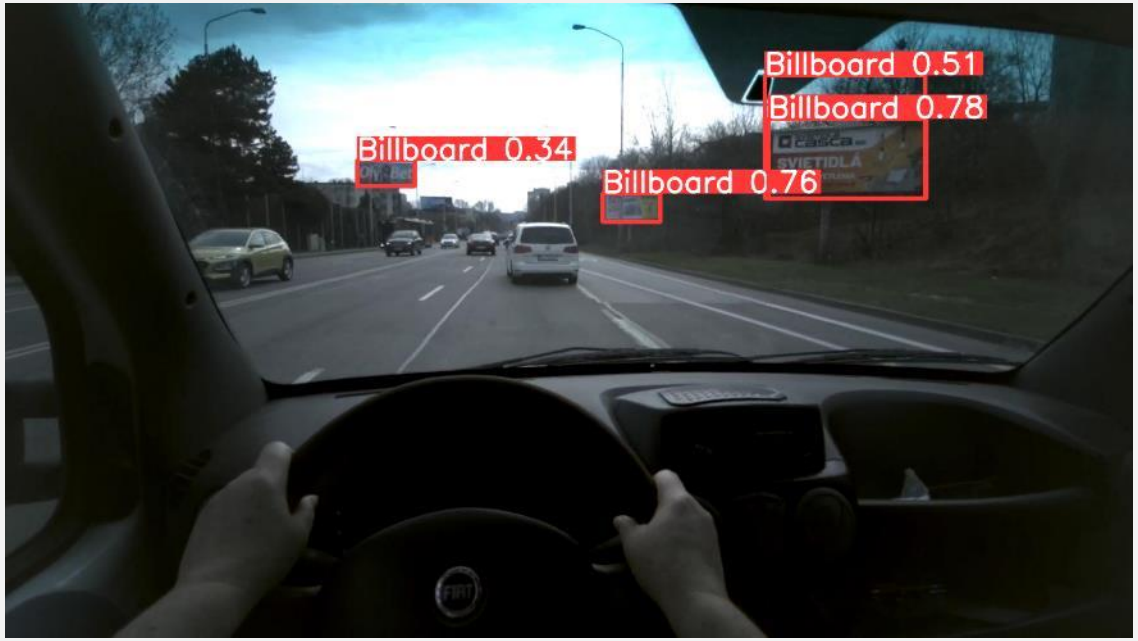
BILLBOARDLAMAC TEST SET



OBJECT DETECTION RESULTS: 2

MAPILLARY VISTAS TEST SET

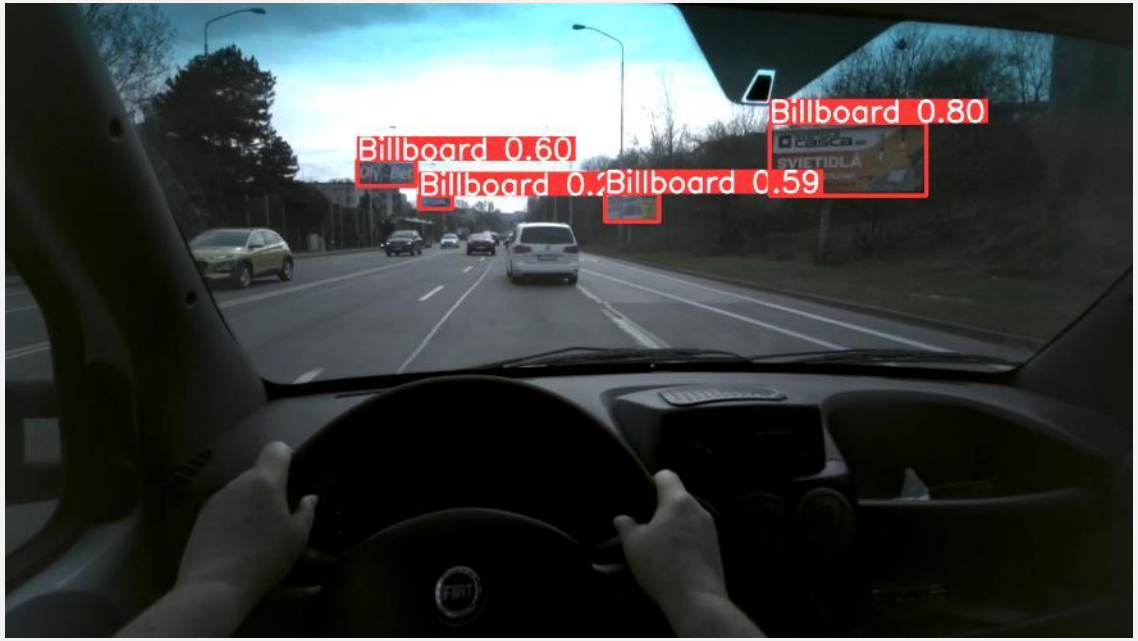
BILLBOARDLAMAC TEST SET



OBJECT DETECTION RESULTS: 3

MAPILLARY VISTAS TEST SET

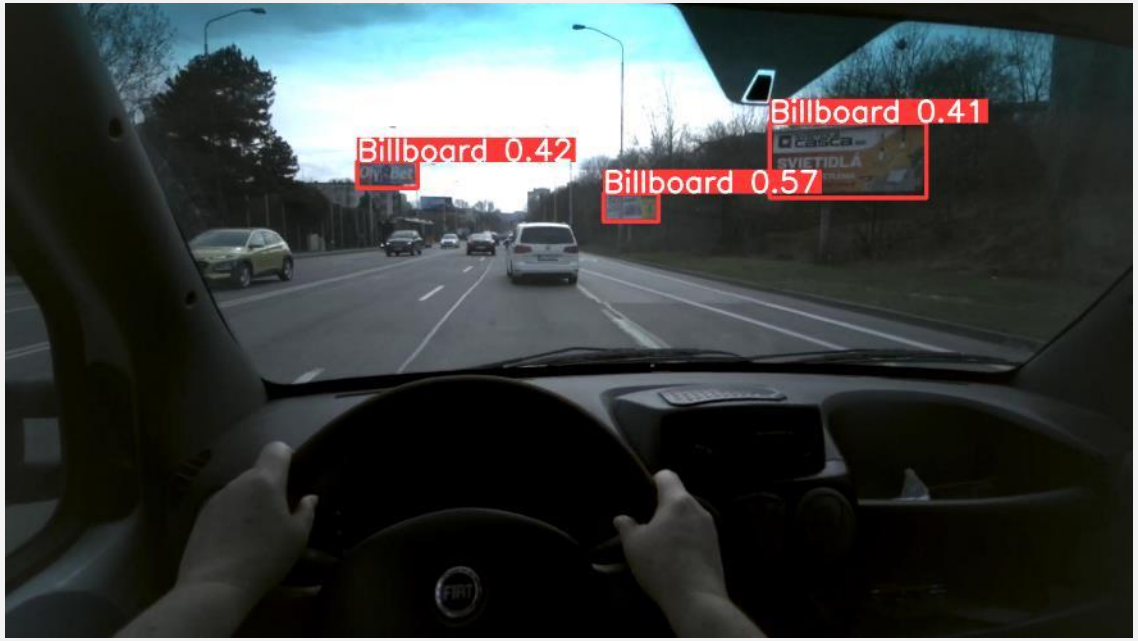
BILLBOARDLAMAC TEST SET



OBJECT DETECTION RESULTS: 4

MAPILLARY VISTAS TEST SET

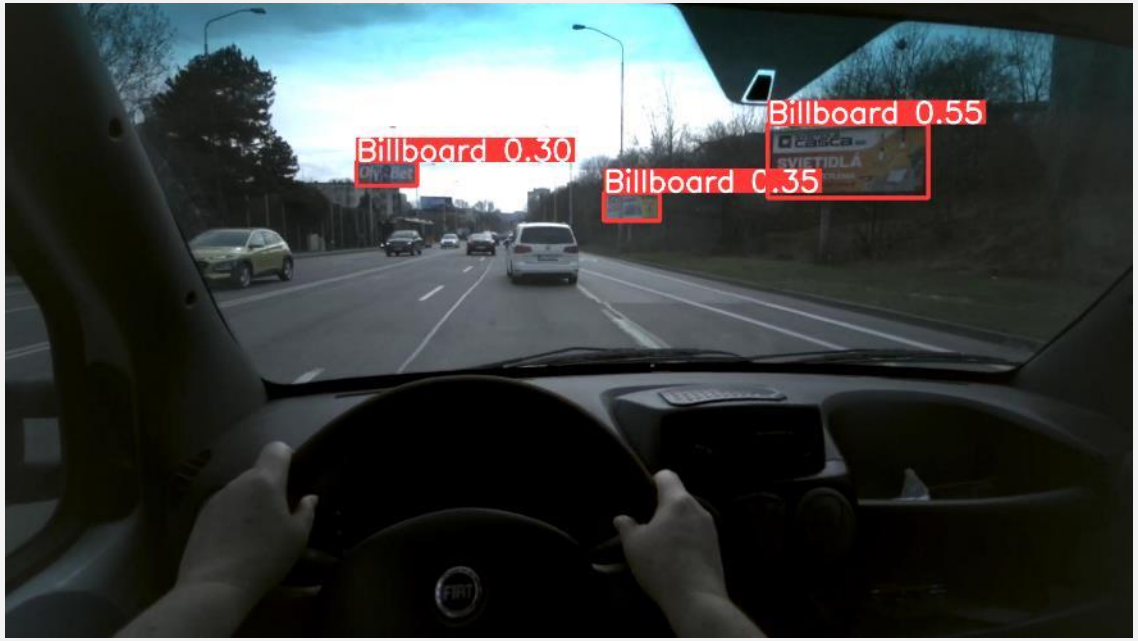
BILLBOARDLAMAC TEST SET



OBJECT DETECTION RESULTS: 5

MAPILLARY VISTAS TEST SET

BILLBOARDLAMAC TEST SET



OBJECT DETECTION RESULTS: 6

```
OutOfMemoryError: CUDA out of memory. Tried to allocate 100.00 MiB
0 GiB reserved in total by PyTorch) If reserved memory is >> alloca
for Memory Management and PYTORCH_CUDA_ALLOC_CONF
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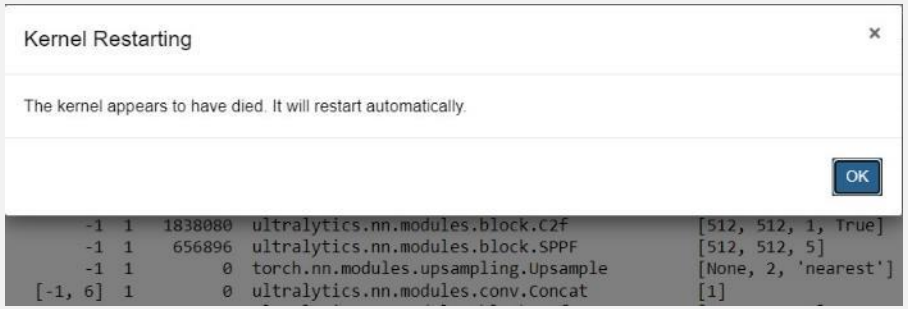
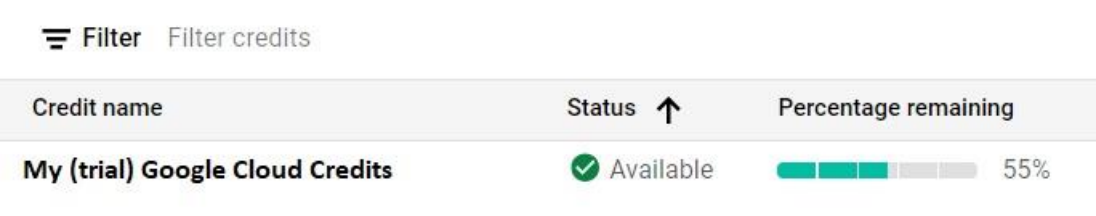
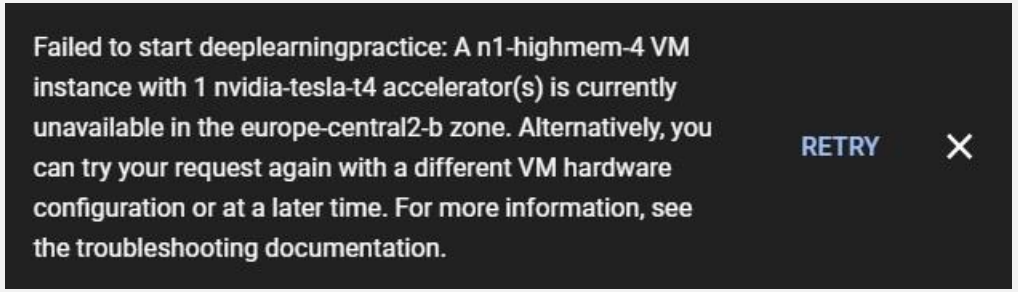
- Previous models used a batch size of 16.
- It is expected that different batch sizes will affect training.
- Google Cloud resources constraints.

CHALLENGES

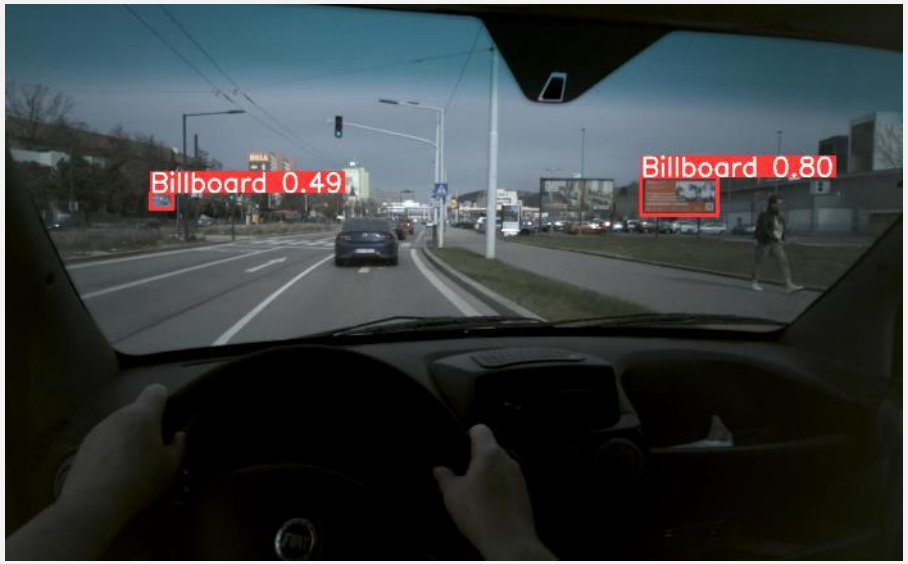


- Difficulties with local training.
- Online resources constraints.
- Availability of online resources.





NEXT STEPS



- Increase dataset size (depends on resources limitation).
- Hyperparameter tunning.
- Build a classifier (deep learning based).
- Assemble the full pipeline.

THANK YOU FOR YOUR ATTENTION

And don’t get distracted while driving!

SCIENTIFIC ARTICLES & GITHUB SITE

Articles:

- https://www.sciencedirect.com/science/article/pii/S0965856418310632
- https://www.sciencedirect.com/science/article/pii/S136984782100245X
- https://www.sciencedirect.com/science/article/pii/S0001457505002149
- https://ieeexplore.ieee.org/abstract/document/10308914
- https://openaccess.thecvf.com/content_iccv_2017/html/Neuhold_The_Mapillary_Vistas_ICCV_2017_paper.html
- https://pure.ulster.ac.uk/en/publications/billboard-detection-in-the-wild

GitHub site:

- https://github.com/carlos-p-t/diploma-thesis